

DTU



Perception for Autonomous Systems 34759:

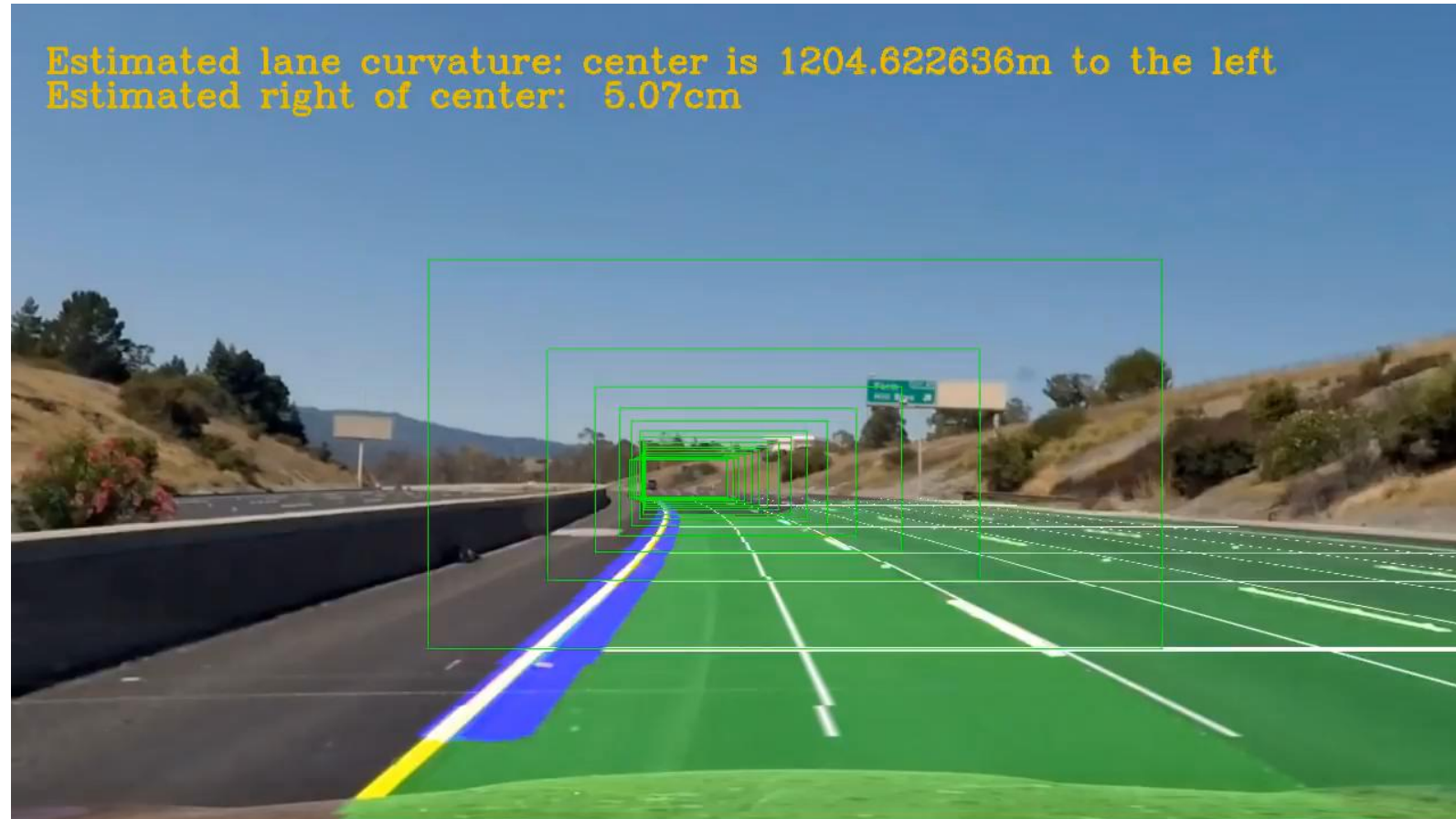
State Estimation - Histogram Filter

Lecturer: Evangelos Boukas—PhD

Cases of State Estimation



Cases of State Estimation



Cases of State Estimation



What is State Estimation

Goal:

- Given a State Vector of a system
- Estimate over time the state using input of external sensors

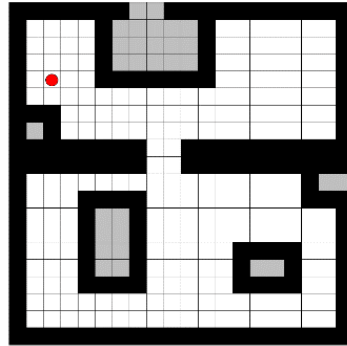
Useful for:

- Localization
- Tracking
- Prediction
- Sensor Fusion
- ...

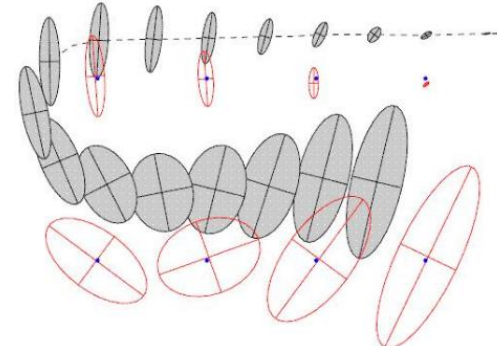
Continuous vs Discrete State

Unimodal vs Multimodal Distribution

- State:

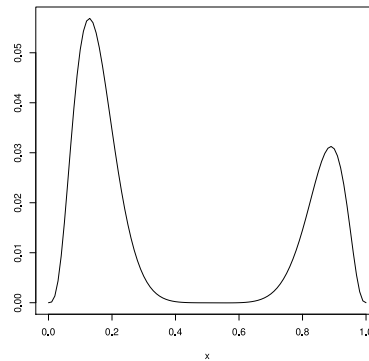


Discrete

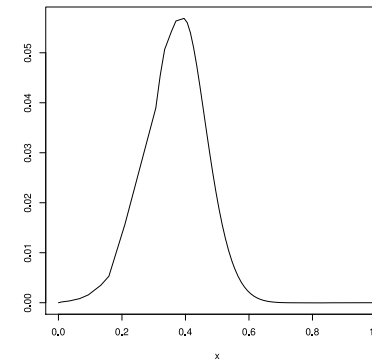


Continuous

- Distribution



Multimodal



Unimodal

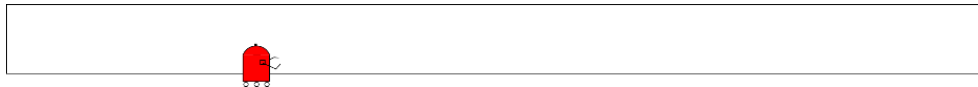
DTU Simple State Estimation Example

- Assume a robot in an area like this one:

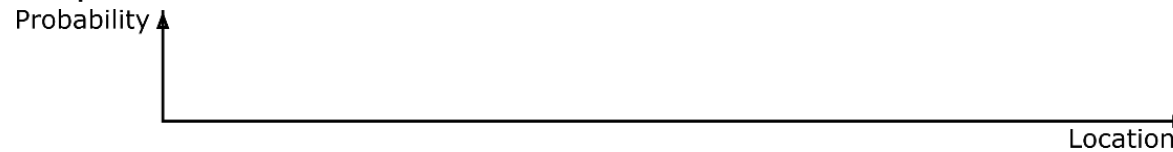


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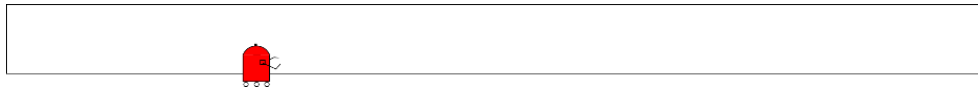


- We model the position of the robot as follows:



DTU Simple State Estimation Example

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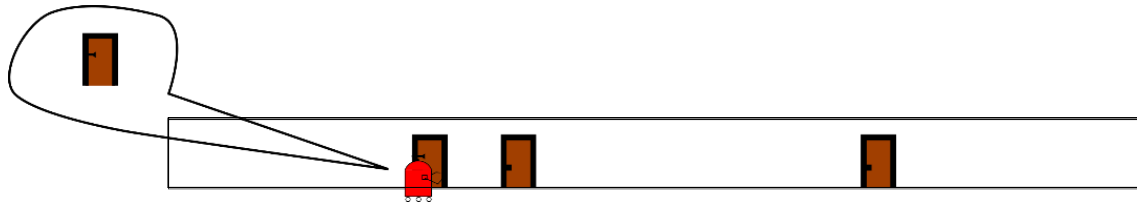


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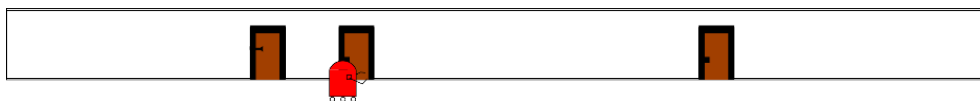


- The robot can distinguish (sense) a door vs “no-door”:

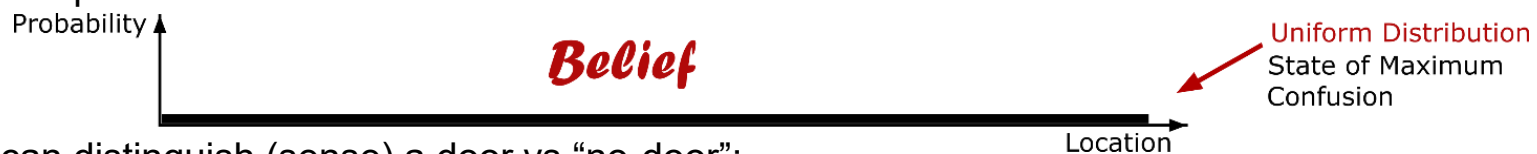


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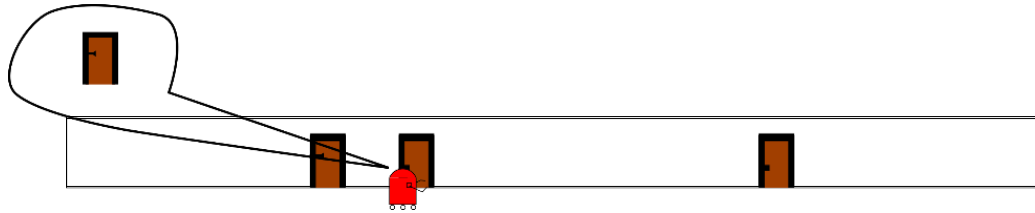


- The robot moves to the right:



DTU Simple State Estimation Example

- Assume a robot in an area like this one:



- We model the position of the robot as follows:



- The robot can distinguish (sense) a door vs "no-door":



- The robot moves to the right:

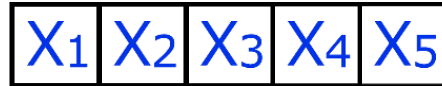


- The robot senses again:



Sense, Let's build it ourselves!

- This state estimation problem is called localization
- Assume a robot which can be in one of five blocks:



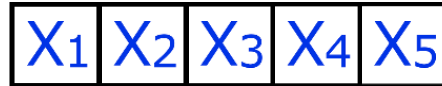
- Without any info, What is the probability of the robot being in each cell?

$$p(X_i) = \frac{1}{5}$$

for i in $(1 \dots 5)$

Sense, Let's build it ourselves!

- This state estimation problem is called localization
- Assume a robot which can be in one of five blocks:



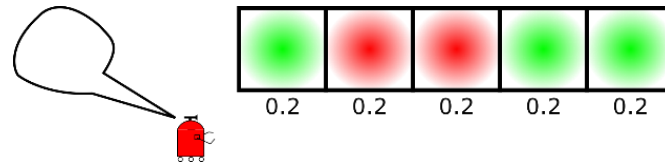
- Without any info, What is the probability of the robot being in each cell?

$$p(X_i) = \frac{0.2}{5}$$

for i in $(1 \dots 5)$

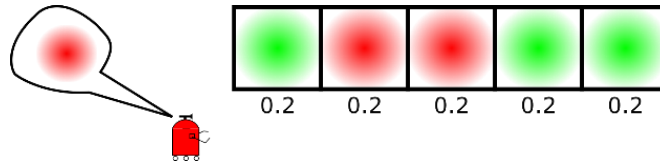
Sense, Let's build it ourselves!

- Now our robot is allowed to sense:

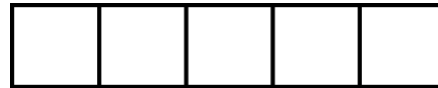


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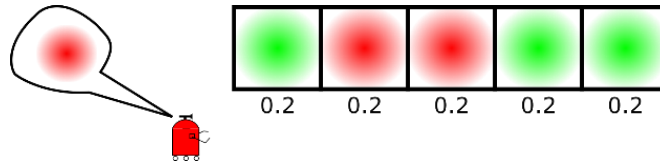
- What does that mean for our probability?



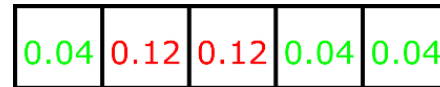
- Let's start by arbitrarily multiplying any correct "sensing" with 0.6 & any incorrect with 0.2

Sense, Let's build it ourselves!

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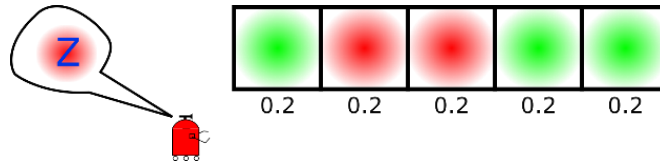
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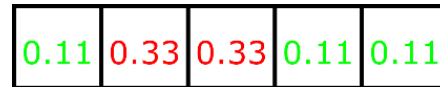
- Let's start by arbitrarily multiplying any correct "sensing" with 0.6 & any incorrect with 0.2
- This is close to the belief. However, it is wrong, why?
- It does not add up to one. How to do it?

Sense, Let's build it ourselves!

- Now our robot is allowed to sense:



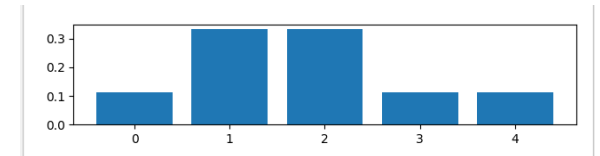
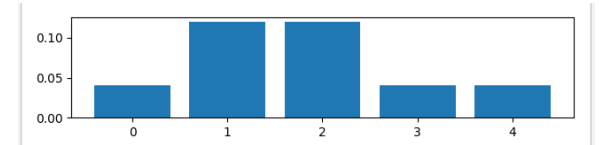
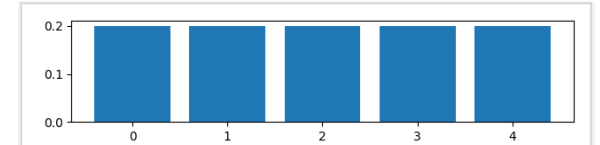
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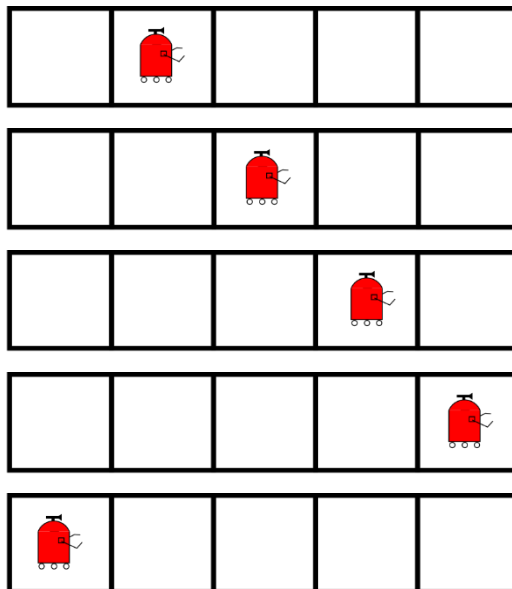
$$p(X_i | Z)$$

Posterior Distribution



Motion, Let's build it ourselves!

- Assume that the space is circular (i.e when moving right in the last cell you go to the first):



Motion, Let's build it ourselves!

- Assume that the space is circular (i.e when moving right in the last cell you go to the first):



- What will happen with the posterior probability?

0.11	0.33	0.33	0.11	0.11
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Motion, Let's build it ourselves!

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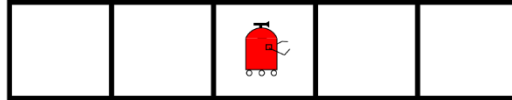


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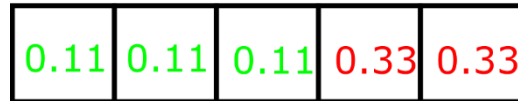


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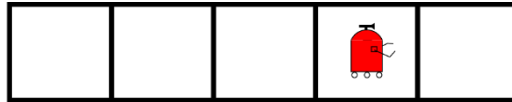


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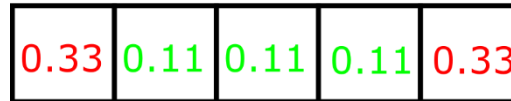


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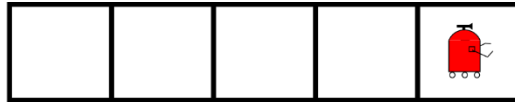


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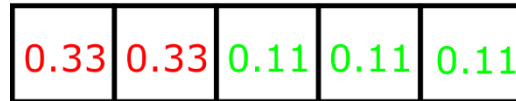


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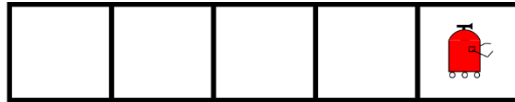
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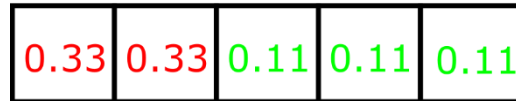
- However, this doesn't happen. Usually the distribution becomes more uncertain:

Motion, Let's build it ourselves!

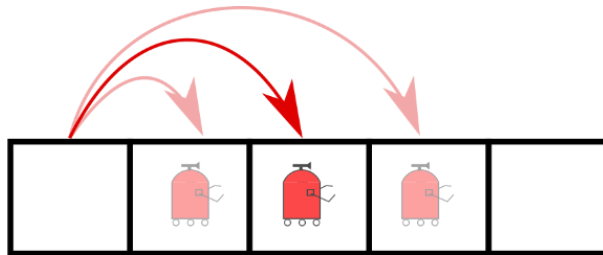
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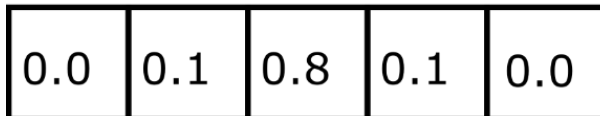


**Uncertain
Motion**

$$p(X_{i+U-1}) = 0.1$$

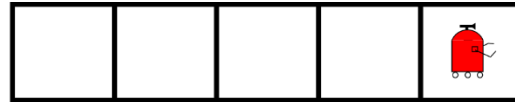
$$p(X_{i+U}) = 0.8$$

$$p(X_{i+U+1}) = 0.1$$

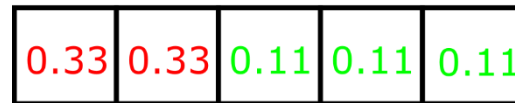


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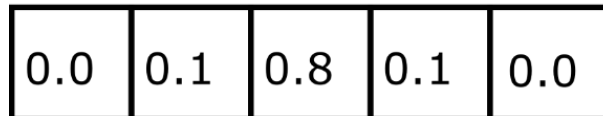
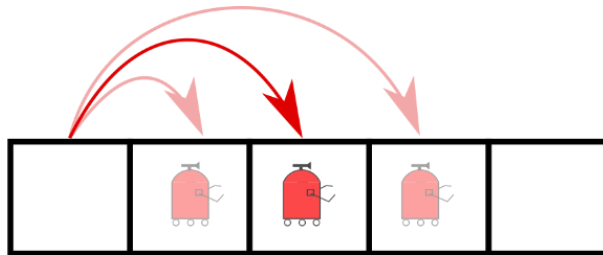
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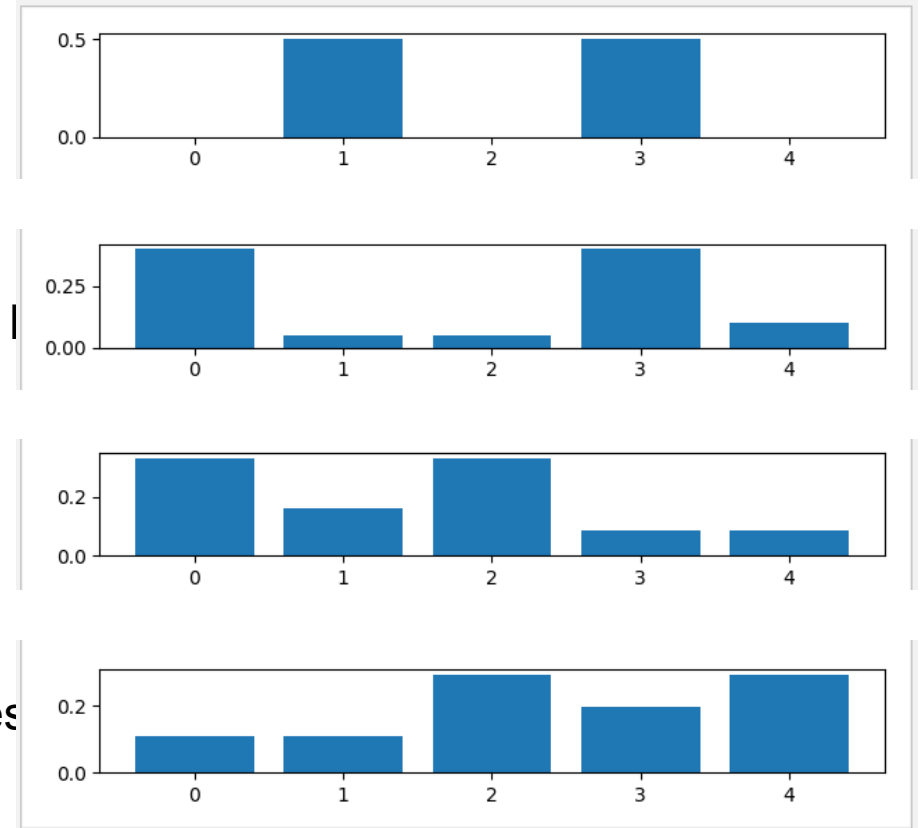


**Uncertain
Motion**

$$p(X_{i+U-1}) = 0.1$$

$$p(X_{i+U}) = 0.8$$

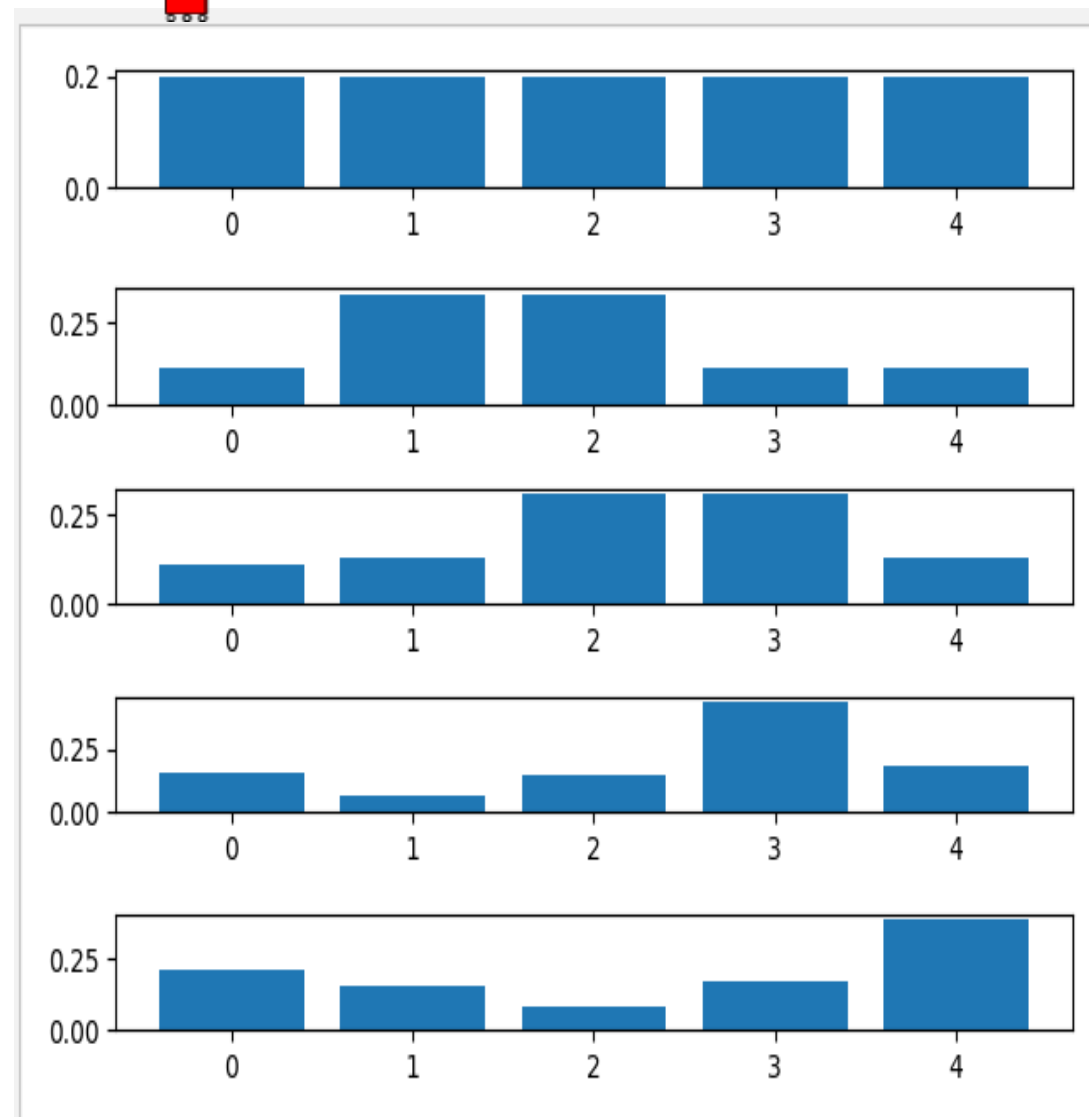
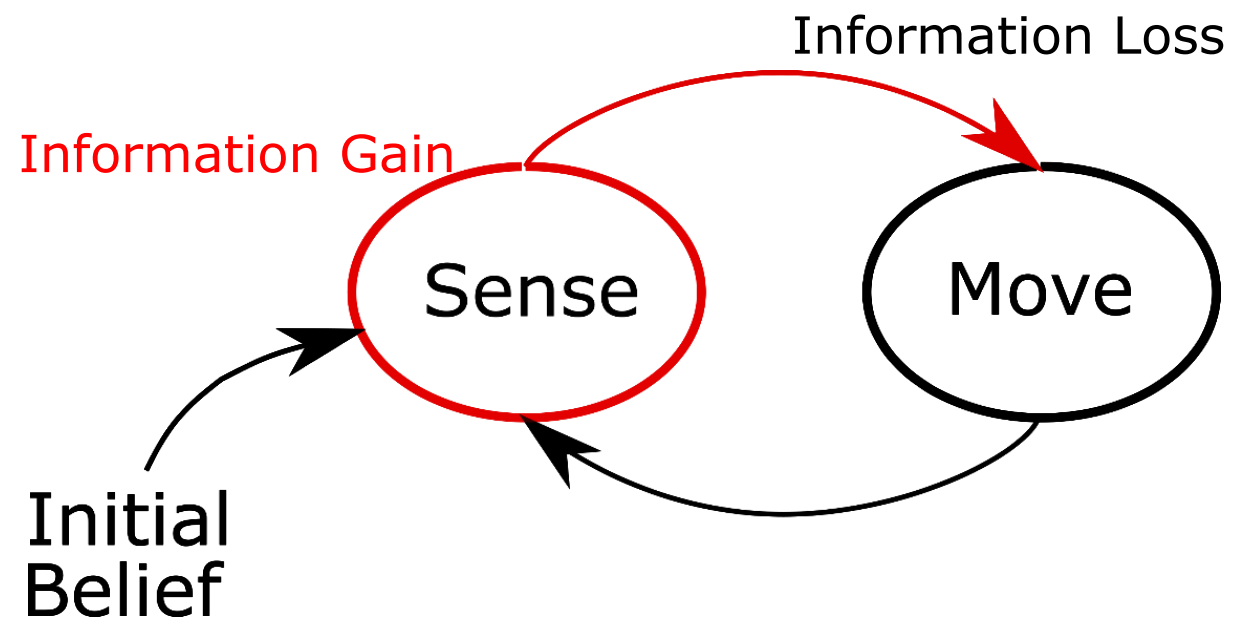
$$p(X_{i+U+1}) = 0.1$$



That's very good!!!!



- Localization is just a sense/move cycle:



Sum Up Global Localization

- Belief  Probability
- Measurements  Multiplication followed by Normalization
- Moving  Convolution

Belief – Formal Definitions

- Probability:

$$0 \leq p(X) \leq 1$$

- Assuming 2 states:

$$p(x_1) = 0.2$$

$$p(x_2) = \underline{\hspace{1cm}}$$

Belief – Formal Definitions

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- Assuming 5 states:

0.1	0.1	0.1	0.1	
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Belief – Formal Definitions

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$$p(x_2) = \underline{0.8}$$

- Assuming 5 states:

0.1	0.1	0.1	0.1	0.6
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Measurement – Formal Definitions

- Bayes Rule
- Assuming a grid cell and the measurements:

X grid cell Z measurement

- The belief of the location given a measurement:

$$p(X_i | Z) = \frac{p(Z | X_i) p(X_i)}{p(Z)}$$

Measurement Probability *Prior*

- A product of the prior with the measurement probability
- The “probability of seeing a measurement independently of location” (normalizer...)

Movement – Formal Definitions

- This is a somewhat complicated formula:
- Notice:
 - Grid Location
 - Time

$$p(\underset{\text{Grid Location}}{X_i^{\overset{\text{Time}}{t}}}) = \sum_j p(\underset{\text{Prior Probability}}{X_j^{t-1}}) \overset{\text{Movement Probability}}{p(X_i | X_j)}$$

- Here are the components:
 - Prior
 - Movement
- This is what is called Total Probability

$$\Pr(A) = \sum_n \Pr(A | B_n) \Pr(B_n),$$

Histogram-based State Estimation

Main histogram-based global localization problem (Markov Localization):

- Memory scaling is exponential
- So, it is unfeasible in large real world problems.

Sum-UP so far

- State Estimation
- Markov Localization
- Probability
- Bayes
- Total Probability

Coming UP:

- Kalman Filter
- Particle Filter???

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